
Symmetries in Physics — Exercise Sheet 13

The Heisenberg group and Fock space

Variant: Questions only

Exercise 1 Realise the Heisenberg group by upper unitriangular matrices $g(a, b, c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$ and the algebra by $X(a, b, c) = \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}$. (i) Show $e^{X(a,b,c)} = g(a, b, c + \frac{1}{2}ab)$. (ii) Verify the exact Baker–Campbell–Hausdorff law by computing both $g(a, b, c)g(a', b', c')$ and the bracket series, and confirm they agree.

Exercise 2 The Weyl form of the canonical commutation relations uses the unitaries $U(a) = e^{-iaP/\hbar}$ and $V(b) = e^{ibQ/\hbar}$ on $L^2(\mathbb{R})$ — translations and modulations — which satisfy $U(a)V(b) = e^{-iab/\hbar}V(b)U(a)$. Show that the Schrödinger representation is irreducible: any bounded T commuting with all $U(a), V(b)$ is a scalar. You may use these point-of-use imports, with the definitions: for a set S of bounded operators, its *commutant* is $S' = \{T : TS = ST \text{ for all } S \in S\}$ and the *von Neumann algebra generated by S* is $(S \cup S^*)''$ — for the family $\{V(b)\}$, closed under adjoints since $V(b)^* = V(-b)$, simply S'' ; an abelian algebra is *maximal abelian* when it equals its own commutant; $L^\infty(\mathbb{R})$ is the space of essentially bounded measurable functions, and M_f is multiplication by f . The imports: the von Neumann algebra generated by the $V(b)$ is $\mathcal{M} = \{M_f : f \in L^\infty(\mathbb{R})\}$, and this multiplication algebra is maximal abelian, $\mathcal{M}' = \mathcal{M}$. In the last step, treat translation invariance as equality of L^∞ classes, not as a pointwise identity.

Exercise 3 Coherent states are $|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$, the eigenvectors $a|\alpha\rangle = \alpha|\alpha\rangle$. (i) Compute ΔQ^2 and ΔP^2 and show the product is the minimum $\hbar^2/4$. (ii) Prove the resolution of identity $\frac{1}{\pi} \int |\alpha\rangle \langle \alpha| d^2\alpha = \mathbb{I}$. (iii) Under $H = \hbar\omega(a^\dagger a + \frac{1}{2})$, show a coherent state stays coherent with $\alpha(t) = \alpha e^{-i\omega t}$, and deduce that $\langle Q \rangle(t), \langle P \rangle(t)$ trace the classical oscillator ellipse. For part (ii), interpret the integral weakly: first test its matrix elements between vectors in $\mathcal{D}_{\text{fin}} = \text{span}\{|n\rangle : n \geq 0\}$. You may use the elementary extension lemma: a bounded sesquilinear form on a dense subspace has a unique bounded extension to the Hilbert space.

Exercise 4 Build the fermionic Fock representation for $d = 2$ explicitly: operators c_1, c_2 with $\{c_i, c_j\} = 0$, $\{c_i, c_j^\dagger\} = \delta_{ij}$, as 4×4 matrices via a Jordan–Wigner construction (Kronecker products of Pauli matrices with a σ_3 string, as in the lecture’s computer-algebra remark).

Exercise 5 For the bosonic Fock space over $\mathbb{C}^d$, let $\mathcal{F}_{\text{alg}} = \bigoplus_{n \geq 0}^{\text{alg}} \text{Sym}^n \mathbb{C}^d$ and define on this core $d\Gamma(A) = \sum_{ij} A_{ij} a_i^\dagger a_j$ for a $d \times d$ matrix A . (i) Show $[d\Gamma(A), a_k^\dagger] = a_{Ae_k}^\dagger$, where $a_v^\dagger = \sum_i v_i a_i^\dagger$ for $v \in \mathbb{C}^d$, and deduce that $d\Gamma$ is a Lie-algebra homomorphism on this common invariant core. (ii) Identify the number operator as $d\Gamma(\mathbb{I})$ and show it is unbounded. For $U \in \text{U}(d)$ define the all-Fock unitary $\Gamma(U) = \bigoplus_n U^{\otimes n}$ on the symmetric sectors and prove $\Gamma(U) a_i^\dagger \Gamma(U)^{-1} = a_{Ue_i}^\dagger$. If A is anti-Hermitian, differentiate at $t = 0$ on $\mathcal{F}_{\text{alg}}$ and state the imported closure formula relating $\Gamma(e^{tA})$ to $d\Gamma(A)$.

Exercise 6 Two identical oscillators coupled by $H = \frac{p_1^2 + p_2^2}{2m} + \frac{1}{2}m\omega^2(x_1^2 + x_2^2) + \frac{1}{2}m\kappa(x_1 - x_2)^2$. Diagonalise by a normal-mode change of variables, give the spectrum, and identify the role of the exchange symmetry S_2 .